

Selfish Player

A Data-Driven Analysis of Shot Decision
Quality Using Expected Goals and Match
Context

https://colab.research.google.com/drive/1jCH6yQHI2UpFtjY2N1rWGd0in9Nxnqzn#scrollTo=eb047JtvH_JZ

Presented by
Zhi You•Zi Feng•Yang

Getting Started



**CAN WE QUANTIFY AND COMPARE PLAYER-LEVEL
SHOOTING DECISION QUALITY USING AGGREGATED
DECISION COST DERIVED FROM MATCH EVENT DATA?**

FROM “SELFISH” TO OPTIMAL: A DATA PERSPECTIVE(DECISION LEVEL)

PROJECT CONTEXT:

This project reframes subjective judgments of “selfish play” into a data science problem by quantifying decision quality using event-level data and contextual features.

HYPOTHESIS:

We hypothesize that decision quality varies systematically across players and match contexts, and that suboptimal shooting decisions can be quantified using expected value differences.



KEY DATA USED IN THIS STUDY



01

Event-Level Data

Shot events (location, xG, time, outcome)
Pass events (start location, timestamp, possession)
Match information (teams, match time, season)

02

Derived Data

Pitch zones based on spatial coordinates
Pass-to-shot sequences within the same possession
Expected goal value of passes (xG_pass)

03

Contextual Information

Match time and scoreline
Team and match context



DATA SOURCE

Dataset: Premier League matches from the past ten seasons

01

StatsBomb Open Data — Event-Level Match Data

<https://github.com/statsbomb/open-data>

02

Football-Data.co.uk
Match Results & Historical Records

<https://www.football-data.co.uk/>

03

League standing

DATA SOURCE SELECTION:

StatsBomb (Event Data)

1. Provides detailed event-level information (shots, passes, xG)
2. Enables decision-level analysis
3. High granularity for feature construction

Football-Data (Match Results)

1. Provides complete and continuous match records
2. Ensures temporal coverage across seasons
3. Complements limited coverage of event data

League Standings (Context Data)

1. Provides team performance context (points, ranking)
2. Captures match importance and pressure
3. Supports contextual feature engineering

HOW WE DID OUR DATA PREPERATION?

Problem Definition & Source Selection → Load Raw Data → Clean and Standardize Tables → Extract Shot/Pass Events → Engineer Context Features → Build Final Analytical Dataset

DATA CLEANING

We first cleaned the result data and standardized match metadata to ensure reliable downstream joins and context construction.

01

Clean football-data results table (cell6)

- Keep only necessary columns
- Unify the date format
- Convert the goals column to numeric
- Remove rows with missing values in key fields

02

Flatten and standardize StatsBomb match metadata (cell9)

We flatten the nested home_team / away_team and extract:

- home_team_name
- away_team_name
- match_date
- season_name

And carry out team name standardization.

Event Extraction & Transformation

from raw nested event JSON to structured shot and pass tables

1. Load event data for all available EPL matches (cell8)
2. Flatten nested event fields (cell 11): The original event JSON from StatsBomb cannot be directly analyzed, so we extracted elements such as `type_name`, `team_name`...
3. Extract shots, passes, and assign pitch zones: Filter out the "shot table" and "pass table" from all the events(cell 12)+ Convert continuous coordinates into discrete pitch zones (cell13)



FEATURE ENGINEERING



1. Estimate passing alternative value (xG_pass): cell14

“If the player had passed instead of shot, what would the historical passing value from that zone look like?”

2. Compare pass value and shot value (delta): cell15

Compare the shot_xg of each shot with the xG_pass of the corresponding zone:

$$\text{delta} = \text{xG_pass} - \text{shot_xg}$$

which is the core of the "decision cost" mentioned later.

3. Add recent-form and league-form context: cell16+17

Make each shot not only a spatial movement, but also incorporate the team's recent form and league ranking context.

4. Add score/context pressure features: cell18+19

$$\text{decision_cost} = \text{delta} \times \text{context_factor}$$



SINCERE APPRRECIATION FOR YOUR TIME AND PATIENCE

Presented by
Zhi You•Zi Feng•Yang

